

Yunus Tuncbilek

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[LinkedIn](#) | [Personal Website](#)

ABOUT ME

Yale CS & Math grad who recognized that a long-standing problem in US Census employment data was really a linear program — the Gurobi solver I built cut per-year runtime from over two days to twenty minutes, replaced an approximate method with exact solutions, and produced a dataset now used across spatial economics. Also shipped an open-source C++/ROS computer-vision wrapper on pre-release social robotics hardware, and spent two years teaching competition-level algorithms.

Returned to software engineering in 2024 after leaving academia in pure mathematics; looking for SDE roles heavy on backend and optimization.

SKILLS

- **Programming Languages:** Python, Java, C++, C, JavaScript, SQL, MATLAB
- **Cloud & Infrastructure:** Google Cloud Platform (GCP), Google Cloud App Engine
- **Frameworks & Databases:** Node.js, Robot Operating System (ROS), OpenCV, Gurobi Optimization Solver, MySQL
- **Core Software Engineering:** Systems Programming, Data Structures & Algorithms (DSA), Object-Oriented Design (OOD), Code Reviews, Version Control (Git)
- **AI & Optimization:** Retrieval-Augmented Generation (RAG), AI System Architecture
- **Technical Leadership:** Engineering Mentorship, Team Leadership (Head TA), Architecture & Design Discussion
- **Advanced Mathematics:** Linear Programming, Optimization, Extremal Graph Theory, Number Theory

HIGHLIGHTED PROJECTS

- [US Census Employment Data Recovery Solver](#)
 - The first dataset to provide fully imputed employment values for every suppressed cell across all years of the CBP (1975–2016), turning a partially-suppressed federal dataset into a complete, research-ready panel now used across spatial and labor economics (+250 academic researcher users). Built with a Gurobi-based linear programming solver run on Yale's HPC cluster.
- [Math-Mate](#) (Solo Project, 2025-2026)
 - An AI tutoring assistant that lets teachers configure the underlying model directly: instructors set custom instructions controlling the AI's tone, pedagogical approach, and how it should guide students toward answers without revealing them outright, giving teachers control over pedagogy that most off-the-shelf AI tutors don't expose. Built with Retrieval-Augmented Generation (RAG) for context-aware, homework-specific guidance.
- Glasshopper
 - Co-founded and led engineering for the first public platform letting Yale students give structured feedback on extracurriculars, alongside one of the first comprehensive directories of all campus organizations, giving students a "knowledge market" to evaluate activities before joining.
 - Built full-stack with Node.js and MySQL, a custom jQuery front-end, and deployed on Google Cloud Platform.
- For a full list of my projects, visit my [personal website](#).

RELEVANT SOFTWARE ENGINEERING EXPERIENCE

Yale University, New Haven, CT, USA

May 2016 – July 2019

Research Assistant - Software Engineering (Full time)

- Identified that a long-standing US Census data-suppression problem – 60% of cells were suppressed for confidentiality – was solvable as a linear problem; built a Gurobi-based Python solver over 5-7M variables per year for 42 years that replaced earlier approximate solvers and recovered all cells exactly, cutting runtime from 2 days to 20 minutes on Yale's HPC clusters.
- Co-created a public dataset spanning 42 years — 150M recovered cell-values — now a standard dataset in spatial and labor economics; the accompanying paper has been cited 250+ times.

Yale University, New Haven, CT, USA

May 2016 – August 2016

Research Assistant - Software Engineering (Part time)

- Engineered a high-performance, open-source ROS wrapper in C++ to integrate a legacy computer vision face-tracker into pre-release social robotics hardware (Jibo).
- Optimized multi-modal data pipelines using linear algebra to translate raw spatial coordinates into real-time face orientation and attention metrics, maximizing data throughput and accuracy.
- Architected low-latency software components for interactive games designed to support cognitive development milestones for children with ADHD, culminating in a successful production launch.
- Maintained clean code standards and version control, publishing the core integration framework as an open-source contribution ([GitHub Repository](#)).

OTHER PROGRAMMING EXPERIENCE

Zhuge Academy Education, Vancouver, BC, Canada

September 2024 – June 2026

Competitive Programming Coach (Full time)

- Promoted to Head of Computer Science for driving instructional excellence in CS; established rigorous code review standards and delivered 6 weekly technical knowledge-sharing sessions that measurably raised overall code quality.
- Architected and delivered comprehensive technical modules on Data Structures and Algorithms along with 200+ LeetCode Medium/Hard Problems to students preparing for Canadian Computing Competitions (CCC), driving nearly all students to advance a full competition level in CCC (from Beginner to Junior or Junior to Senior).
- Leveraged data-driven methodologies to teach 10+ concurrent technical courses as sole instructor, achieving a 100% retention rate in Fall 2025 through effective technical communication and project-based milestones.
- [GitHub Repository](#) of my coding exercises for the students.

EDUCATION

University of Wisconsin-Madison, WI, USA

Graduate Studies in Mathematics (PhD Program)

September 2019 – July 2023

GPA: 3.750

- Passed all doctoral qualifying examinations required for PhD Candidacy.
- Completed 64 credits (22 courses) of advanced graduate courses for research in Analytic Number Theory.

Master of Arts in Mathematics

December 2022

GPA: 3.625

Yale University, New Haven, CT, USA

Bachelor of Science in Computer Science and Mathematics

September 2014 – May 2018

GPA: 3.83

- Distinction in the Major (Earned A or A- in three-quarters of the credits in the major subject)
- Earned *cum laude* (Top 30 percent of the graduating class in GPA)

AWARDS

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| • Top 500, Putnam Mathematics Competition, USA | 2017 |
| • Gold Medal, Mediterranean Mathematics Olympiads and Silver Medal, National Olympiads | 2014 |
| • First Place, National Mathematics Research Competition of Turkey | 2013 |

RESEARCH & PUBLICATIONS

Carnegie Mellon University, Pittsburgh, PA, USA

May 2017 – July 2017

Research Assistant | Summer Undergraduate Applied Mathematics Institute

- Selected for an elite summer research institute focused on extremal graph theory and discrete mathematics.
- Co-authored and published peer-reviewed research titled "*Anti-Ramsey Multiplicities*" ([arXiv:1801.00474](#)), solving open problems regarding edge-coloring properties in complete graphs.

UNIVERSITY TEACHING EXPERIENCE

University of Wisconsin-Madison, WI, USA

August 2019 – May 2023

Graduate Teaching Assistant

- Earned the Exceptional Teaching Award after serving as the Sole Instructor for two math courses.

- Facilitated technical discussion sections for CS 435 (Introduction to Modern Cryptography); focused on the transition from theoretical primitives to provably secure architectures.